

XRPL Validator — Daily Engineering Report • 2026-04-01

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 5 commits (xrpl-validator 5)

Engineering changes

- **perf: flat 256-bucket parallel hash + recursive delete collapse fix** (ee3356e , xrpl-validator)
- Replace SHAMap tree with flat sorted array split into 256 buckets - Rayon-parallel compute across 256 buckets: 2.5s total (0.4s merge + 2.1s hash) - spawn_blocking keeps SSE feed smooth during hash computation - Fix SHAMap delete: recursive structural collapse prevents inner → inner → leaf chains - Single-leaf collapse in InnerNode::hash() for correct hash at any tree depth - Gap ledgers use fast path (cache merge ...)
- **perf: 256-bucket dirty tracking + rippled migration to the reference node** (3439731 , xrpl-validator) Replace full-array merge with per-bucket binary search updates (0.65s → 0.03s). Only recompute dirty bucket hashes via rayon (skip ~30% clean buckets). Update RPC endpoints from the validator host localhost to the reference node ([host]).
- **feat: network resilience — RPC failover + endpoint migration** (c90dde8 , xrpl-validator)
- rippled_client.rs: update primary endpoint to the reference node ([host]) - bulk_sync.rs: replace hardcoded RPC_URL with rpc_failover() that cycles through local → xrplcluster → s1.ripple.com - All source files: migrate hardcoded [host] → [host] - rippled now runs on the reference node (NVMe, systemd, 3.1.2) - Validator on the validator host has full CPU with no rippled contention
- **feat: persist leaf cache to disk for instant restarts** (160db5e , xrpl-validator)
- FlatHasher::save(): atomic write (tmp + rename) of 1.2GB binary cache - FlatHasher::load(): reads cache + recomputes bucket hashes in 2.5s - Auto-save every 500 matches in background thread - SIGTERM handler saves cache before exit - Cache path set next to RocksDB (leaf_cache.bin) - Restart time: 30s → 2.5s (12x faster)

Docs and operations

- **docs: update README to reflect current validator status** (81295d1 , xrpl-validator)

Appendix — commits

Time	Repo	Commit	Subject
00:47	xrpl-validator	ee3356e	perf: flat 256-bucket parallel hash + recursive delete collapse fix
18:33	xrpl-validator	81295d1	docs: update README to reflect current validator status

Time	Repo	Commit	Subject
20:37	xrpl-validator	3439731	perf: 256-bucket dirty tracking + rippled migration to the reference node
20:59	xrpl-validator	c90dde8	feat: network resilience — RPC failover + endpoint migration
21:40	xrpl-validator	160db5e	feat: persist leaf cache to disk for instant restarts